

Quiz (Carolina Reaper)

- Q1.** A shipping company has 2^5 trucks. If each truck can carry 3^2 packages, how many packages can they carry in total?
- (A) $2^5 \cdot 3^2 = 288$
(B) $2^5 \cdot 3^2 = 324$
(C) $2^5 \cdot 3^2 = 486$
(D) $2^5 \cdot 3^2 = 972$
(E) $2^5 \cdot 3^2 = 1458$
- Q2.** Simplify the expression: $(2^3)^4$
- (A) $2^{3 \cdot 4} = 2^{12}$
(B) $2^{3+4} = 2^7$
(C) $2^{3-4} = 2^{-1}$
(D) $2^{3 \cdot 4} = 2^6$
(E) $2^{4-3} = 2^1$
- Q3.** A traffic flow model uses the equation $d = 2t^3$ to estimate the distance traveled. If $t = 2^2$, what is the distance traveled?
- (A) $d = 2(2^2)^3 = 2 \cdot 2^6 = 128$
(B) $d = 2(2^2)^3 = 2 \cdot 2^4 = 32$
(C) $d = 2(2^2)^3 = 2 \cdot 2^2 = 8$
(D) $d = 2(2^2)^2 = 2 \cdot 2^4 = 32$
(E) $d = 2(2^2)^2 = 2 \cdot 2^2 = 8$
- Q4.** Simplify the expression: $2^5 \cdot 2^3$
- (A) $2^{5-3} = 2^2$
(B) $2^{5+3} = 2^8$
(C) $2^{5 \cdot 3} = 2^{15}$
(D) $2^{3-5} = 2^{-2}$
(E) $2^{5/3}$
- Q5.** A delivery company has 3^4 drivers. If each driver can deliver 2^3 packages, how many packages can they deliver in total?
- (A) $3^4 \cdot 2^3 = 648$
(B) $3^4 \cdot 2^3 = 864$
(C) $3^4 \cdot 2^3 = 972$
(D) $3^4 \cdot 2^3 = 1296$
(E) $3^4 \cdot 2^3 = 1458$
- Q6.** Simplify the expression: $(3^2)^3$
- (A) $3^{2 \cdot 3} = 3^6$
(B) $3^{2+3} = 3^5$
(C) $3^{2-3} = 3^{-1}$
(D) $3^{2 \cdot 2} = 3^4$
(E) $3^{3-2} = 3^1$
- Q7.** A transportation company uses the equation $t = 2^d$ to estimate the time taken for a journey. If $d = 3^2$, what is the time taken?
- (A) $t = 2^{3^2} = 2^9$
(B) $t = 2^{3^2} = 2^6$
(C) $t = 2^{3^2} = 2^3$
(D) $t = 2(3^2) = 2 \cdot 9 = 18$
(E) $t = 2(3^2) = 2 \cdot 3 = 6$
- Q8.** Simplify the expression: $3^6 \cdot 3^2$
- (A) $3^{6-2} = 3^4$
(B) $3^{6+2} = 3^8$
(C) $3^{6 \cdot 2} = 3^{12}$
(D) $3^{2-6} = 3^{-4}$
(E) $3^{6/2} = 3^3$

Open Ended Questions (FRQ)

- Q9.** A logistics company uses the equation $c = 2^d \cdot 3^t$ to estimate the cost of shipping. If $d = 2^3$ and $t = 3^2$, what is the cost of shipping? Show all steps and simplify your answer.
- Q10.** A traffic flow model uses the equation $v = 3^t \cdot 2^d$ to estimate the velocity of traffic. If $t = 2^2$ and $d = 3^2$, what is the velocity of traffic? Show all steps and simplify your answer.

Chef's Tips

- Ensure students use the correct exponent rules for simplification
- Check for common mistakes such as incorrect application of exponent rules
- Remind students to show all steps and simplify their answers